Piano playing in quiet and in pink noise: a single-subject fNIRS pilot (LYDIA)

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1 Methods

1.1 Participant, task and sessions

A student pianist (sub-01, graduate level) played Gymnopédie no. 1 (Erik Satie) on a piano in a music recording studio hall. In the NOISE condition a loudspeaker placed behind and to the left of the player emitted pink noise. In QUIET the room was silent. Sessions were run in the order `quiet_1` (ses-08), `noise_1` (ses-09), `quiet_2` (ses-10), `noise_2` (ses-11), all on 2026-09-07. Each session was one run of 6 play blocks (planned 40 ± 8 s, jittered) separated by rest (30 ± 5 s), preceded by a 60 s baseline. A 1 kHz beep cued each play (1 beep) and rest (2 beeps) onset. Markers were sent over the Lab Streaming Layer (LSL) with codes 20 = play(QUIET), 21 = play(NOISE) and 30 = rest. Executed play blocks lasted 32–46 s (mean 38.4 s, `data/blocks.csv`); the design of the four runs is shown in Fig. 1.

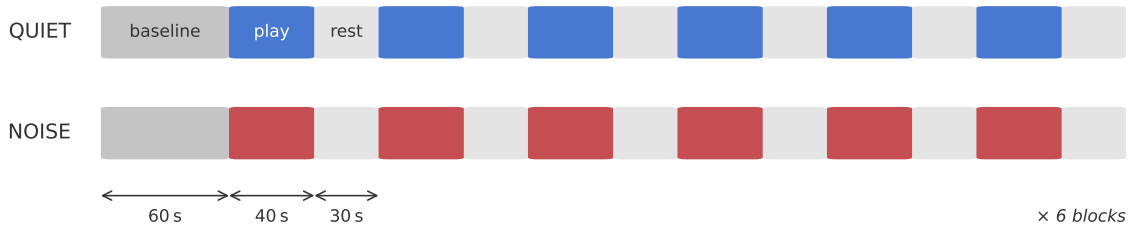


Figure 1: Block design (schematic). Each run began with a 60 s baseline and alternated 6 play blocks with rest; the QUIET and NOISE runs share the same structure. Play and rest durations were jittered around the nominal 40 and 30 s shown here (executed play blocks 32–46 s), and the GLM modelled every block as a 40 s boxcar.

1.2 Montage and acquisition

Functional near-infrared spectroscopy (fNIRS) data were acquired with a NIRSport2 (Aurora 2025.2) at 760/850 nm and 12.2 Hz. The optode layout (Fig. 2) uses 16 sources and 16 detectors on EEG 10-10 sites, giving 47 channels of 30–42 mm separation (mean 35.4 mm). Coverage: frontal pole and ventrolateral prefrontal cortex (VLPFC; Fpz, AF7, AF8 → Fp1, Fp2, AFz), dorsolateral prefrontal cortex (DLPFC) (AF3, AF4, Fz, F3, F4 → F5, F1, F2, F6), premotor and supplementary motor area (SMA; FC5, FC1, FC2, FC6 → FC3, FCz, FC4), sensorimotor (C3, C4 → C5, C6) and temporal (T7, T8 → FT7, TP7, FT8, TP8).

1.3 Quality control

Signal quality was assessed per run in NIRWizard 0.2.1 in non-overlapping 5 s windows using the scalp coupling index (SCI, cardiac band 0.5–2.5 Hz) and peak spectral power (PSP). A window was

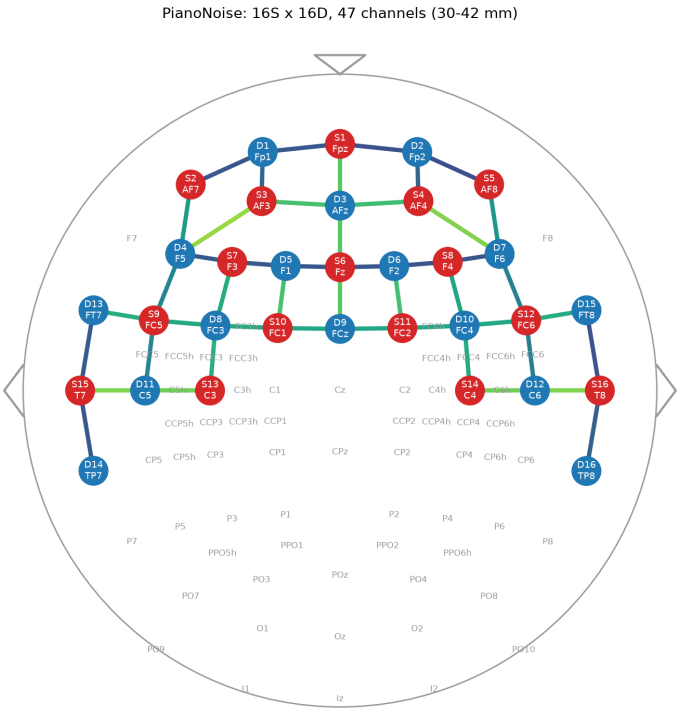


Figure 2: The optode layout, top view (nose up). Red: sources S1–S16, blue: detectors D1–D16, lines: the 47 channels (colour = source–detector distance, 30–42 mm).

good when SCI ≥ 0.8 and PSP ≥ 0.1, and a channel was good when at least 75 % of its windows were good and SNR ≥ 2. The census is reported descriptively (Table 1); no channels were excluded, and all 47 channels enter the analyses below.

1.4 Regions of interest

Ten regions of interest (ROIs) were defined in NIRWizard as five left/right pairs: VLPFC, DLPFC, temporal, premotor (PMA) and SMA (Fig. 3, Table 2). Some channels belong to two ROIs (e.g. S1-D3 and S6-D3 to both DLPFC ROIs, S6-D9 to both SMA ROIs, S9-D4 to left VLPFC and left PMA, S12-D7 to right VLPFC and right PMA).

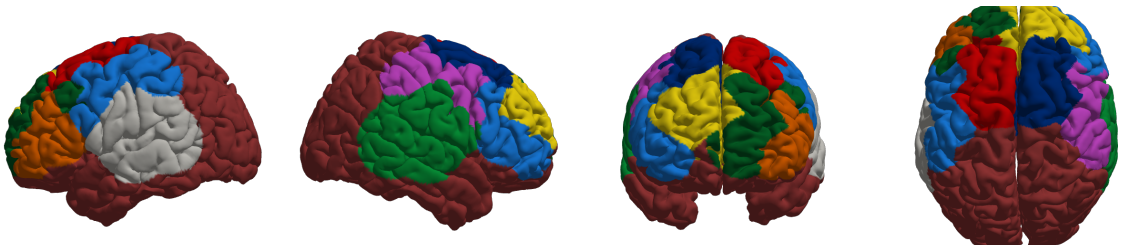


Figure 3: The ten ROIs on the cortex (left, right, anterior, superior view).

1.5 Preprocessing

Preprocessing was done in NIRWizard 0.2.1 (cedalion 26.5.1 backend): (1) conversion of raw intensity to optical density; (2) motion correction in three stages, recursive PCA (9 artefact segments, 15 components removed), temporal derivative distribution repair (TDDR) and wavelet correction (db2, level 4, IQR 1.5); (3) zero-phase Butterworth band-pass 0.01–0.2 Hz; (4) modified Beer–Lambert law with age-dependent differential pathlength factors (assumed age 25) to ΔHbO and ΔHbR in

µM. The montage has no short-separation channels (all separations ≥ 30 mm), so no short-channel regression was applied and scalp and systemic signals remain in the data.

1.6 Analysis

Block averages. Computed per run. Epochs of -10 to 50 s around each play onset were baseline-corrected by the mean of the -10 to 0 s window and averaged over the 12 blocks per condition (6 per run, no epoch rejection). ROI curves are the mean over the channels of each ROI.

GLM. For the GLM the four raw recordings were first concatenated into a single session (26 159 samples) and then preprocessed together with the pipeline above, so that motion correction, filtering and the concentration conversion act on one coherent signal rather than on four separately scaled ones. The design held one regressor per condition (12 blocks each, 40 s boxcar convolved with the canonical SPM HRF) with its temporal derivative as nuisance, an intercept and three run intercepts. It was fitted per channel and chromophore with AR-IRLS: BIC-selected autoregressive prewhitening, pooled over the session and restarted at each run boundary, with robust iterative reweighting. Each condition is tested against the rest baseline. p-values were corrected with Benjamini–Hochberg FDR across channels within condition $\times$ chromophore (residual degrees of freedom 26 151). Maps are shown unthresholded and thresholded, at $|t| > 20$ for HbO and, because the HbR response is smaller, at $|t| > 10$ for HbR.

Laterality. The Seghier index $\mathrm{LI} = (L - R)/(|L| + |R|)$ was computed per ROI pair from the ROI mean GLM β (and t), with $|\mathrm{LI}| > 0.2$ read as lateralised. LI is interpretable when both sides are positive.

2 Results

2.1 Quality control

Run	Duration (s)	Bad / 47	SCI fail	PSP fail	GVTD median
quiet_1	622	28	11	26	0.0095
noise_1	516	26	13	24	0.0099
quiet_2	498	23	13	22	0.0104
noise_2	507	29	11	27	0.0107

Table 1: Per-run QC census. GVTD: global variance of the temporal derivative.

Between 23 and 29 of 47 channels fell below the PSP criterion in each run, while SNR never failed and GVTD was low and stable across runs (medians 0.0095–0.0107), so the recordings are clean of motion but the cardiac signal is weak in part of the montage. The weaker channels lie over the premotor, lateral SMA and temporal optodes (S7–S15), consistent with hair; prefrontal channels were good throughout. Results from the PMA and SMA ROIs should be read with this in mind.

2.2 Block averages

In both conditions HbO rose at play onset in the prefrontal, temporal and premotor ROIs, peaked 7–9 s after onset, fell back to roughly a third of the peak by 15 s and stayed slightly elevated to the end of the block (Fig. 4, Fig. 5). The response is an onset transient followed by a low plateau. HbR changed little (within about ± 0.05 µM) and did not show the decrease that accompanies a typical neurovascular response; in the left-hemisphere ROIs it rose slightly. The time course has nearly the same shape in every prefrontal and temporal ROI, which suggests a systemic contribution alongside cortical activation. The SMA ROIs showed small responses (< 0.15 µM). The QUIET and NOISE

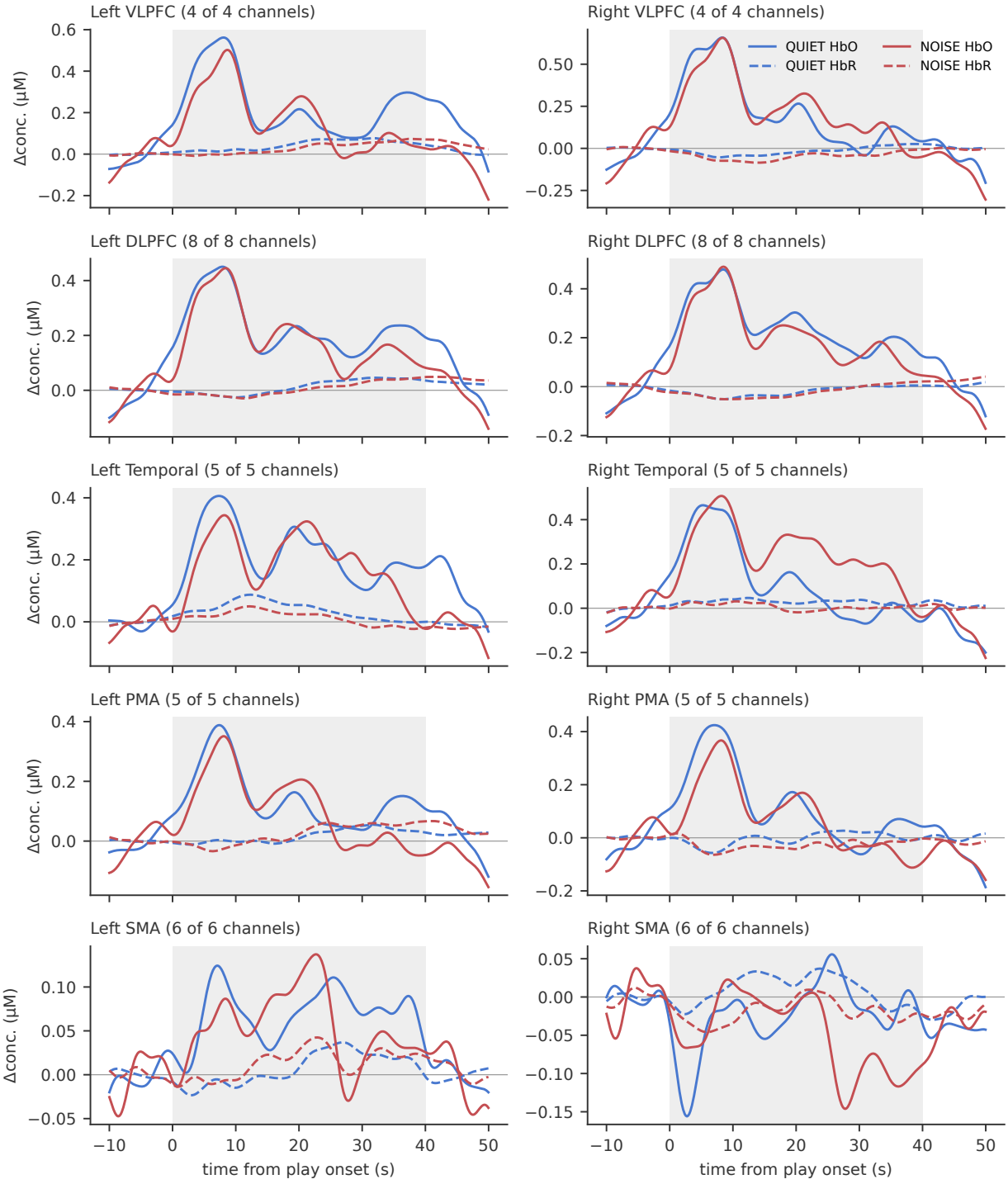


Figure 4: ROI block averages (12 blocks per condition). Solid: HbO, dashed: HbR. Grey: nominal 40 s play block. Note the different y scales.

ROI	Ch.	QUIET HbO	QUIET HbR	NOISE HbO	NOISE HbR
Left VLPFC	4/4	0.230	0.048	0.165	0.034
Right VLPFC	4/4	0.197	-0.016	0.229	-0.049
Left DLPFC	8/8	0.226	0.016	0.194	0.007
Right DLPFC	8/8	0.245	-0.019	0.208	-0.021
Left Temporal	5/5	0.224	0.038	0.200	0.013
Right Temporal	5/5	0.103	0.029	0.252	0.006
Left PMA	5/5	0.140	0.021	0.108	0.028
Right PMA	5/5	0.114	-0.001	0.076	-0.030
Left SMA	6/6	0.077	0.011	0.059	0.016
Right SMA	6/6	-0.013	0.014	-0.048	-0.018

Table 2: Mean block-averaged concentration change 5–40 s after play onset (µM). Ch.: channels in the ROI.

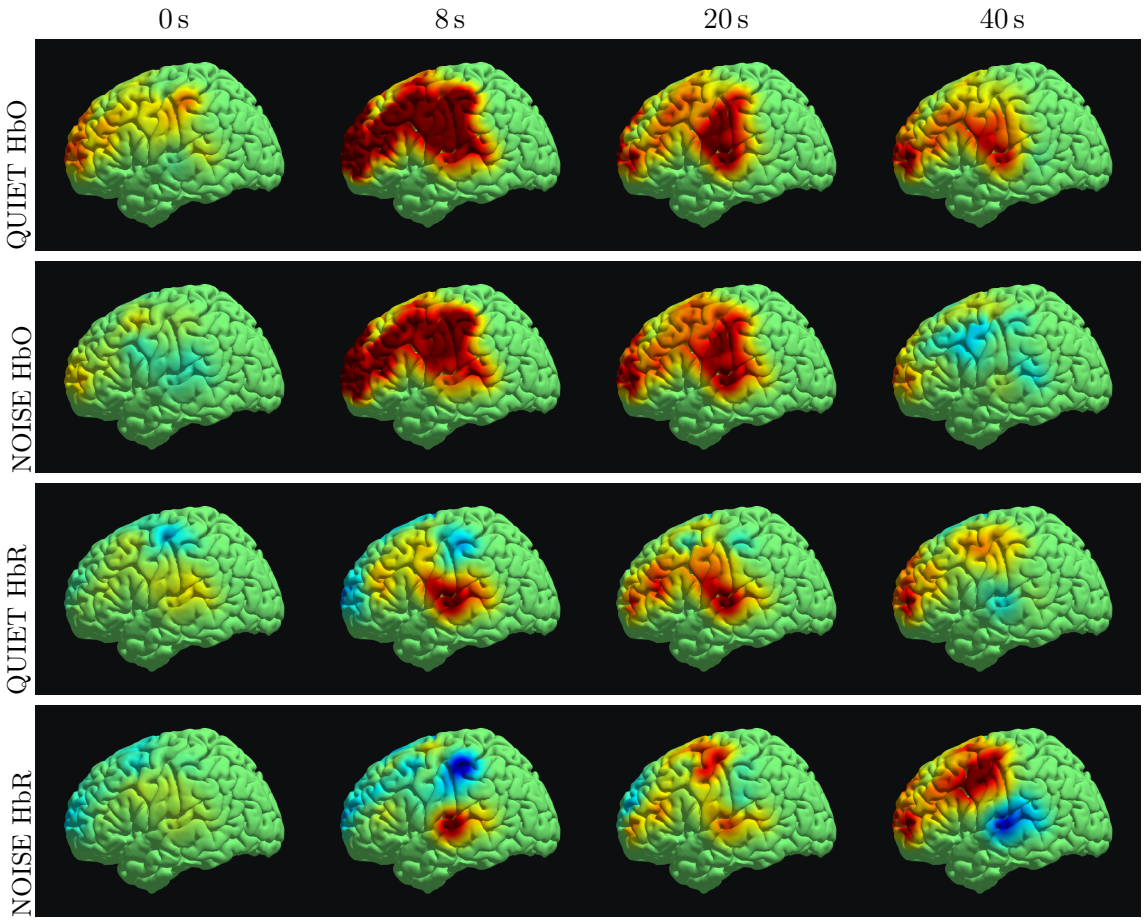


Figure 5: Block-averaged ΔHbO (top two rows) and ΔHbR (bottom two rows) on the cortex, left view, at four times after play onset. Frames from the movies in `analysis/04-block-average/hbo-movies` and `hbr-movies`.

curves largely overlap; the plateau was higher in NOISE in the right temporal ROI (0.25 vs 0.10 µM) and higher in QUIET in left VLPFC (0.23 vs 0.17 µM).

2.3 GLM

Condition	Signal	$q < .05$, $t > 0$	$q < .05$, $t < 0$	$\lvert t \rvert > 20$	median β	t range
QUIET	HbO	45	1	15	0.131	-2.9 to 45.0
QUIET	HbR	25	16	2	0.004	-22.3 to 20.9
NOISE	HbO	42	2	11	0.108	-17.7 to 37.8
NOISE	HbR	21	18	3	0.000	-25.3 to 23.2

Table 3: Channel-level GLM results over the 47 channels: number of channels with FDR $q < 0.05$ by sign, number with $|t| > 20$, median β (µM) and range of t.

	QUIET HbO			NOISE HbO		QUIET HbR		NOISE HbR	
ROI	n	β	t	β	t	β	t	β	t
Left VLPFC	4	0.229	20.7	0.261	23.4	0.023	12.8	0.016	7.6
Right VLPFC	4	0.111	11.0	0.104	9.9	0.002	1.6	-0.005	-1.7
Left DLPFC	8	0.195	21.4	0.140	16.4	-0.004	-2.4	0.001	2.3
Right DLPFC	8	0.182	17.7	0.092	8.0	-0.014	-7.5	-0.014	-8.7
Left Temporal	5	0.181	25.6	0.170	23.0	0.029	12.9	0.010	3.3
Right Temporal	5	0.146	16.5	0.163	19.5	0.022	6.9	0.010	3.1
Left PMA	5	0.100	14.5	0.117	16.9	0.008	3.0	0.003	0.9
Right PMA	5	0.153	17.6	0.079	9.0	-0.009	-2.3	-0.035	-9.7
Left SMA	6	0.067	11.9	0.057	10.7	0.008	3.7	0.017	10.2
Right SMA	6	0.041	6.9	-0.018	-2.6	0.001	-0.2	-0.002	-1.0

Table 4: ROI mean GLM β (µM) and mean t over the n channels of each ROI.

HbO β was positive and significant in 45 of 47 channels for QUIET and 42 of 47 for NOISE (Table 3, Fig. 6). At $|t| > 20$ (Fig. 7), 15 channels remained in QUIET and 11 in NOISE. In QUIET these were left temporal (S15-D11, $t = 45$; S9-D11, S9-D13), medial and left DLPFC (S1-D3, S1-D1, S6-D3, S7-D4), left VLPFC (S2-D1, S2-D4), right DLPFC (S4-D3, S4-D7) and single premotor and right temporal channels. In NOISE the left VLPFC, left temporal and medial DLPFC channels were again the strongest, joined by right temporal (S12-D12, $t = 38$; S12-D15, S16-D15), while the right DLPFC channels dropped below the threshold. By ROI (Table 4), responses were largest in left VLPFC (0.23 and 0.26 µM in QUIET and NOISE), DLPFC and the temporal ROIs and smallest in the SMA. The right-hemisphere DLPFC and PMA responses were about half as large in NOISE as in QUIET (DLPFC 0.18 vs 0.09, PMA 0.15 vs 0.08 µM), whereas the left-hemisphere and temporal responses were similar in the two conditions.

HbR β was an order of magnitude smaller, centred on zero, and significant in both directions in similar numbers of channels (Fig. 6, bottom rows); 13 channels in QUIET and 15 in NOISE exceeded $|t| > 10$ (Fig. 7). It increased in left VLPFC and the temporal ROIs and decreased in right DLPFC and, in NOISE, right PMA; the HbO-up, HbR-down pattern of a typical haemodynamic response was thus clearest in right DLPFC. The t values are large because the fit has 26 151 residual degrees of freedom from one person's 12 Hz samples: they express how consistently the signal follows the block timing within this session.

2.4 Laterality

HbO responses were left-dominant overall (whole-hemisphere LI 0.10 in QUIET, 0.29 in NOISE; Fig. 8). VLPFC was left-lateralised in both conditions (0.35 and 0.43). DLPFC was bilateral in QUIET (0.03) and left-lateralised in NOISE (0.21), because the right-hemisphere response was

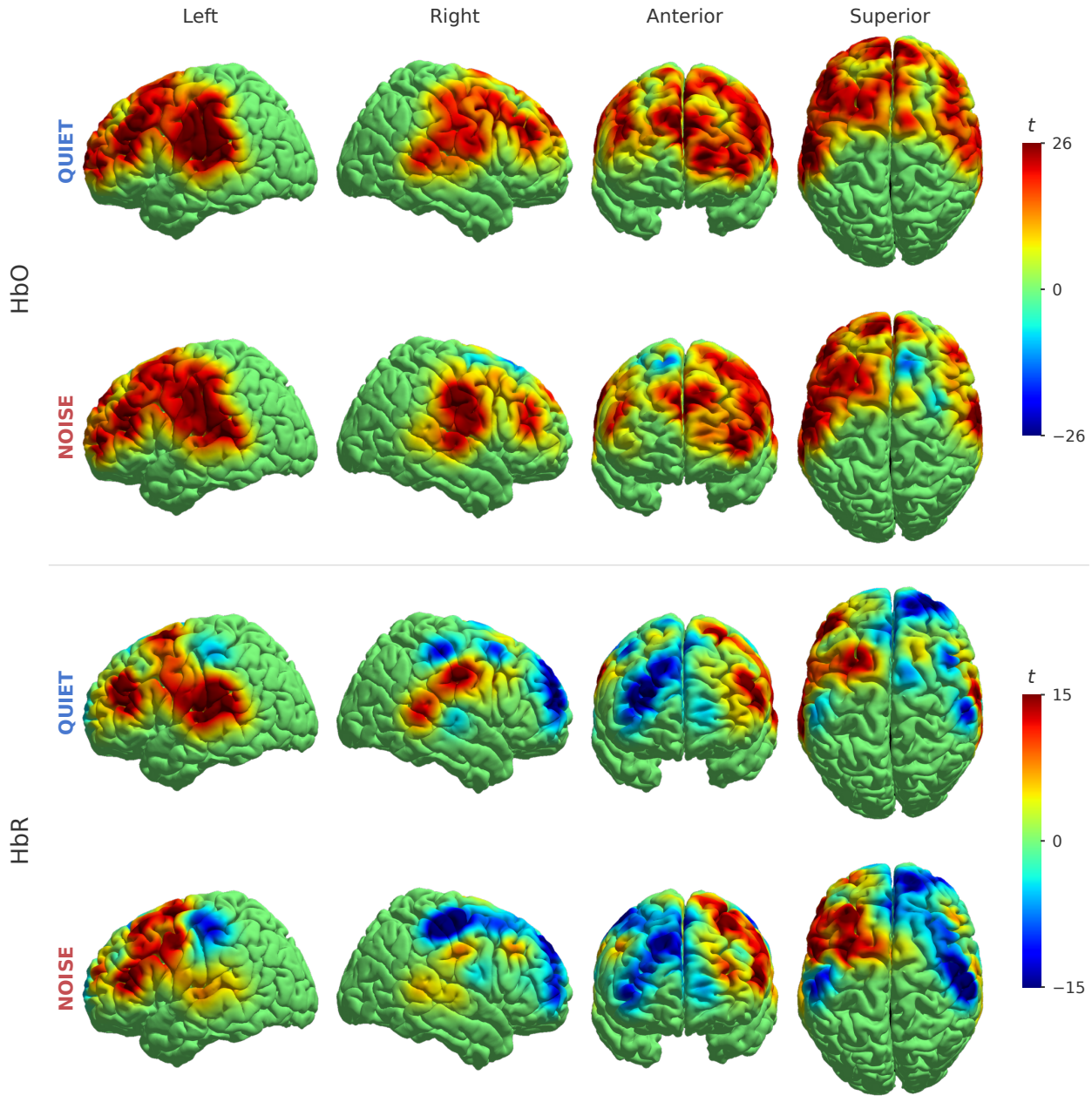


Figure 6: GLM t maps on the cortex, unthresholded, for HbO (colour scale ± 26) and HbR (± 15). For HbR, positive t is an increase.

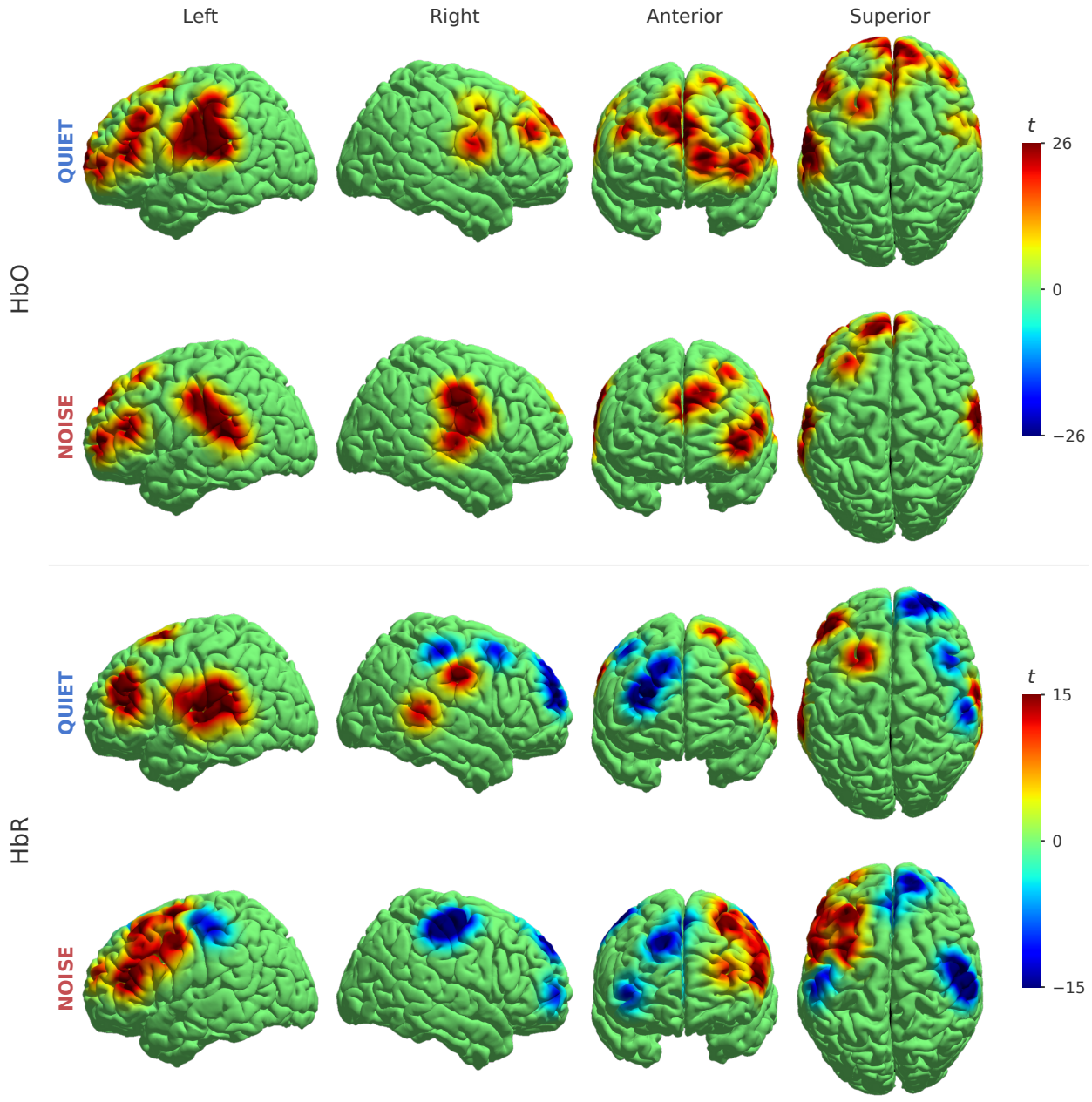


Figure 7: GLM t maps thresholded at $|t| > 20$ for HbO and $|t| > 10$ for HbR, layout and colour scales as Fig. 6.

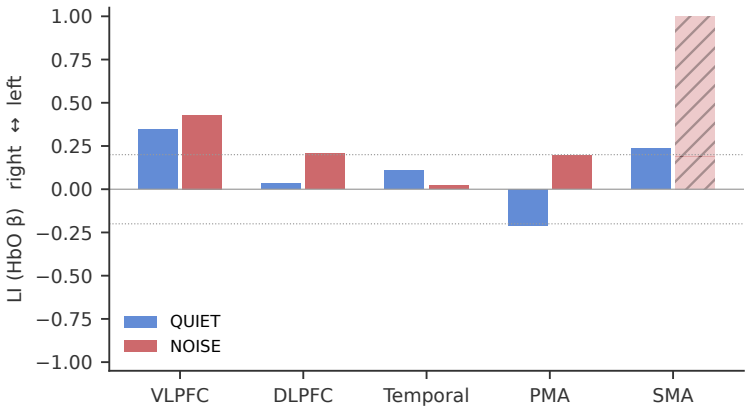


Figure 8: Laterality index of the HbO GLM β per ROI pair. Dotted lines: ± 0.2. Hatched: left and right have opposite signs, so $|\mathrm{LI}| = 1$ regardless of magnitude.

smaller in NOISE while the left changed less. The temporal ROIs were bilateral in both conditions (0.11, 0.02). PMA was right-lateralised in QUIET (-0.21) and close to left-lateralised in NOISE (0.19). SMA was left-lateralised in QUIET (0.24); its value of 1.0 in NOISE reflects a negative right-SMA β and is not a meaningful magnitude. Indices from t values agree in sign and ordering. HbR indices are not reported: the HbR amplitudes are near zero and of mixed sign, which makes the ratio unstable.

3 Discussion

Piano playing produced a robust HbO increase over prefrontal, temporal and premotor cortex in both conditions, strongest in left VLPFC, DLPFC and the temporal ROIs, and left-dominant overall. The response was an onset transient followed by a low plateau, similar in shape across regions, and HbR did not decrease outside right DLPFC. This points to a systemic contribution, which this montage cannot remove because it has no short-separation channels.

The two conditions gave similar maps. Descriptively, the right DLPFC and right PMA responded less in NOISE, which made the NOISE pattern more left-lateralised, and the right temporal response was somewhat larger in NOISE. These are observations in one participant and one session.